



Aquifer recharge: unpacking the impact of deep roots and land use

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Abstract

Deep-rooted trees in native forests and silvicultural systems strongly influence evapotranspiration (ET) dynamics and aquifer recharge. This study synthesizes current scientific evidence on how land use change and plant root system characteristics affect evapotranspiration and groundwater recharge in tropical aquifers. Based on an integrative review of the international scientific literature, 3,671 publications were analyzed, of which 116 were selected according to thematic relevance and methodological quality. The selected studies were organized into two main themes: (i) impacts of land use on the water balance, and (ii) the influence of deep-rooted vegetation on ET dynamics and recharge. Results indicate that forests often reduce recharge relative to pastures, particularly in humid climates (10–30%) and semi-arid regions (70%), mainly due to higher evapotranspiration and root water uptake. However, recent studies highlight hydraulic redistribution and enhanced vertical soil connectivity promoted by root-induced macropores may increase infiltration and recharge even under high-ET conditions. Agroforestry systems and conservation practices, including no-till, also improve soil structure, moisture retention, and infiltration. Thus, although classical literature often associates forests with reduced recharge, recent evidence reveals contrasting outcomes, including reports of spring recovery in reforested areas. Recharge magnitude depends on soil properties, vegetation type, climate, management, and root traits, indicating that forest–recharge relationships are strongly context dependent. These findings have direct implications for sustainable land use planning and groundwater resource management in tropical environments.

Keywords Groundwater · Evapotranspiration · LULC change · Ecohydrology · Water balance

Introduction

The role of groundwater is far beyond being an important water resource. Within the hydrological cycle, aquifer discharge is essential for the maintenance and sustainability of rivers and surface water bodies, a process controlled by recharge and influenced by land use, geology, geomorphology, and climate, as well as their temporal variability (Mengistu et al. 2022; Scanlon et al. 2005).

Land use and land cover (LULC) influence recharge in unconfined aquifers and are commonly assessed using

watershed-scale water balance techniques. Notably, however, these methods have not adequately evaluated the withdrawal of water from aquifer storage by vegetation, when the water table is deeper (>5 m), as occurs under forests in tropical and temperate climates. Land-use changes such as urbanization, agricultural expansion, and deforestation can significantly alter infiltration processes, leading to either reductions or increases in aquifer recharge over time (Gebreslassie et al. 2025; Zhang and Schilling 2006).

Scanlon et al. (2007) conducted a comprehensive analysis of land-use change impacts on groundwater recharge in the Colorado River Basin, United States, highlighting that conversion of natural ecosystems to non-irrigated intensive agriculture reduced natural recharge to near zero. In contrast, recent studies report that increased forest cover may also reduce recharge due to the higher evapotranspiration compared to agricultural or pasture systems (Li et al. 2023; Gherardi and Sala 2020; de Ferraz et al. 2019; Wu et al. 2019; Jackson et al. 2005).

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Studies on the influence of rooting depth on groundwater recharge indicate that deep-rooted vegetation can access long-residence-time groundwater storage (> 50 years) (Li et al. 2023; Zhang et al. 2017; Miguez-Macho and Fan 2021). Thus, the interannual soil water balance assessments used in forestry planning, as part of water uptake, may be derived from recharge that occurred decades earlier, making estimates of aquifer storage and renewal rates uncertain (de Ferraz et al. 2019; Lima 2011). Consequently, neglecting water extraction from depths greater than 5 m in water balance calculations may result in underestimation of evapotranspiration and overestimation of groundwater recharge. Under such conditions, vegetation may also exhibit physiological adaptations, such as reduced leaf area, to limit water demand (Zhang et al. 2019), as there is less surface area available for transpiration.

This study aims to synthesize existing scientific evidence on how land use change and plant root structure influence evapotranspiration and groundwater recharge in tropical aquifers. Through an integrative literature analysis, the study compiles and compares reported evapotranspiration and recharge values across land-use types, with particular emphasis on the role of deep-rooted vegetation. The analysis further seeks to identify recurring patterns, methodological limitations, and knowledge gaps, contributing to a more process-based understanding of aquifer recharge under tropical conditions.

Methodology

Groundwater recharge is a well-established concept widely addressed in the literature. In contrast, the impacts of vegetation characteristics, particularly deep-rooted systems typical of forests, on evapotranspiration and recharge remain less explored and insufficiently quantified. Accordingly, this study employed a structured integrative review of the scientific literature addressing the influence of vegetation characteristics, especially root depth, on aquifer recharge processes. The review focuses on the interactions among evapotranspiration, infiltration, and percolation under different land use and climatic conditions. A graphical workflow of the literature search and study selection process is presented in Fig. 1.

The literature search was conducted in early 2025 using SciELO and ScienceDirect databases. The retrieved studies covered the publication period from 1965 to 2024. The same search strategy was applied in both databases: (“recharge” AND “evapotranspiration” AND (“root zone” OR “macropores”)). Only peer-reviewed journal articles published in English and Portuguese were considered.

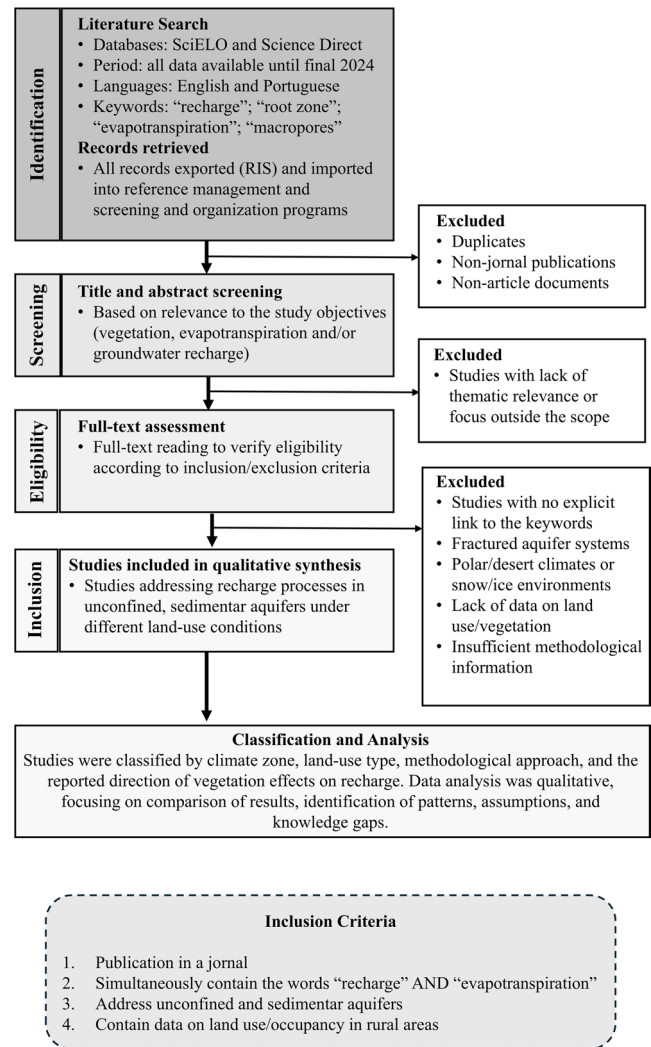


Fig. 1 Workflow of article selection and thematic screening

Studies were included when they explicitly addressed at least one of the following aspects: (i) the role of vegetation or root depth in controlling evapotranspiration or percolation; (ii) quantitative or qualitative estimates of groundwater recharge under different land use or vegetation types; or (iii) conceptual or modeling approaches linking vegetation processes to groundwater recharge. Studies focusing exclusively on surface hydrology without subsurface implications were excluded, as were those addressing fractured aquifer systems. Publications focused on polar, or desert, snowy, or ice-dominated climates, as well as conference abstracts and institutional documents, were also excluded.

Rather than applying a formal systematic review protocol, the selected studies were analyzed using a thematic and process-based framework. Each article was classified by climate zone, land-use type, methodological approach (field measurements, modeling, remote sensing, or conceptual

analysis), and the reported direction of vegetation effects on recharge. This approach enabled the identification of recurrent patterns, dominant assumptions, and methodological limitations across contrasting environmental settings.

All retrieved records were exported in Research Information Systems (RIS) format and imported into Mendeley for reference management and Rayyan for thematic screening and organization. The selection process involved an initial screening of titles and abstracts, followed by full-text assessment to ensure relevance to the study objectives. The analysis focused on recharge processes in unconfined sedimentary aquifers under different land-use conditions, particularly in rural and agricultural settings. The following inclusion criteria were applied for study selection:

1. Publication in a journal;
2. Simultaneously and necessarily contain the words “recharge” and “evapotranspiration”;
3. Reference free and sedimentary aquifers;
4. Contain data on land use and occupancy, including variations in their use, with focus on rural areas.

Based on their thematic focus, the selected studies were categorized into two main groups: (i) effects of land use change on groundwater recharge and (ii) influence of deep-rooted vegetation on evapotranspiration dynamics. Data analysis was qualitative to semi-quantitative, emphasizing comparison of reported results, identification of recurring patterns, and recognition of knowledge gaps. Summary ranges and averages presented in the tables were compiled from the subset of reviewed studies providing comparable numerical information, including 97 studies with precipitation data by land use and climate, 40 studies with recharge data by land use, 34 with evapotranspiration (ET) values, 16 with ET/recharge ratios, 37 with recharge/precipitation ratios, and 34 with ET/precipitation ratios. No formal quantitative meta-analysis, effect-size weighting, or statistical pooling

of results was performed. Instead, the review adopted a qualitative-to-semi-quantitative comparative synthesis focused on recurring trends, contrasts among environments, and reported directional responses in recharge. As with any literature-based review, the dataset may reflect uneven regional research intensity and publication bias. Therefore, some identified patterns should be interpreted with caution, particularly in underrepresented regions.

Results

Literature selection and temporal distribution

The initial search yielded 3,671 articles published between 1965 and 2024 in English and Portuguese (Fig. 2). After title and abstract screening, 893 references were retained for further evaluation. A subsequent full-text assessment for thematic relevance yielded a final dataset of 116 studies.

From the earliest publications on the subject until the early 2000s, annual output remained low, generally ranging from one to four studies per year. From 2015 onward, publication frequency increased markedly, peaking between 2020 and 2021, reflecting the advances in hydrological modeling, growing concerns over water security under increasing water scarcity, and the expansion of cultivated areas.

Figure 3 presents the geographic distribution of the studies and the main land-use types investigated. The process sought to ensure global representativeness, methodological diversity, and a critical approach to the topic of recharge versus land use.

Most studies combined historical, current, and projected climate data with numerical flow modeling, sensitivity analyses, and scenario simulations under different land use and climate-change conditions. The reviewed literature showed a strong predominance of studies conducted in rural settings, particularly those involving agriculture, planted forests, and eucalyptus plantations.

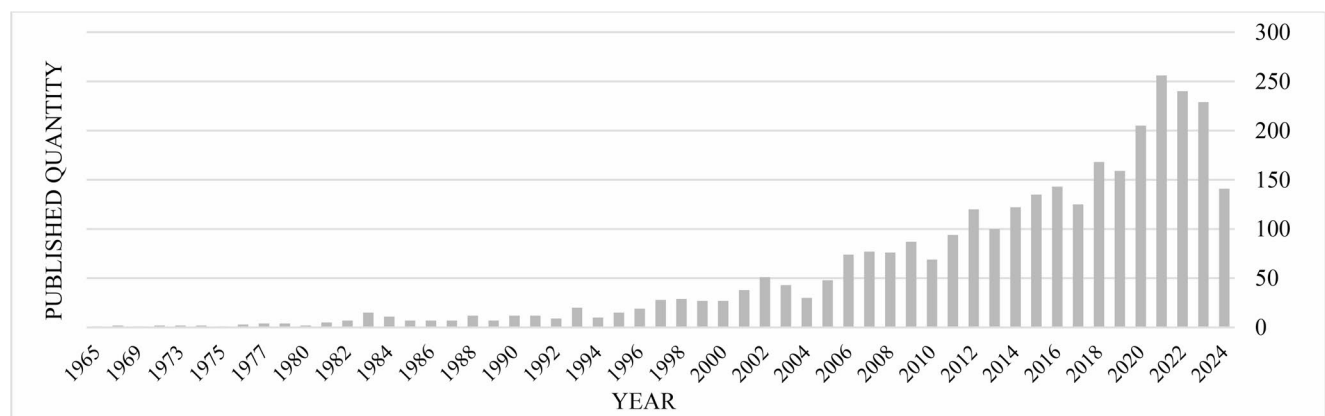
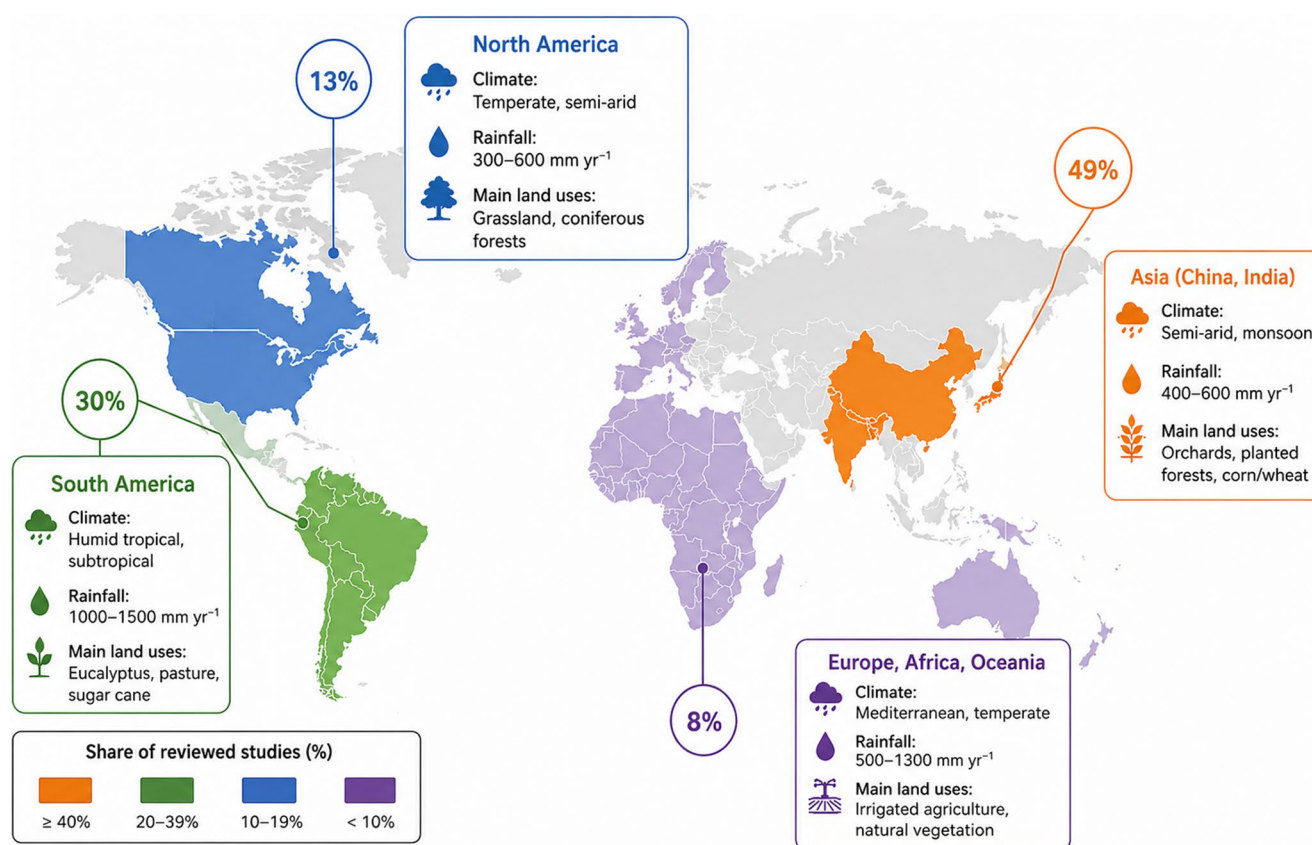


Fig. 2 Number of publications on the relationship between recharge and land use and occupation



Percentages indicate the proportion of studies ($n = 92$) conducted in each region according to the reviewed literature.

Fig. 3 Global distribution of reviewed studies and dominant environmental settings

Physical controls of aquifer recharge

The reviewed studies cover four major climatic zones: humid/subtropical, tropical, semi-arid/monsoonal, and temperate/semi-arid. Each zone presents marked variations in precipitation regimes and predominant land uses, which directly influence recharge processes. The results of this review show that aquifer recharge is strongly modulated by a combination of biophysical factors.

Impact of vegetation and land use

Studies conducted in Brazil found that eucalyptus plantations can reduce recharge by up to 80% compared to pasture areas (Mattos et al. 2019; Lucas and Wendland 2016; Wendland et al. 2007), an effect attributed to high water demand and rainfall interception by the canopy.

In Latin America, aquifer recharge studies focus on agricultural regions or threatened ecosystems, where land use strongly influences water flows. In Brazil, research has been mainly conducted in the states of São Paulo, Minas Gerais, and Bahia (Oliveira et al. 2005a, b, 2019; Mattos et al. 2019; de Ferraz et al. 2019; Christina et al. 2017; Lucas

and Wendland 2016; Hubbard et al. 2010; Stape et al. 2004), where mean annual rainfall is approximately 1,445 mm. Reported recharge values range from less than 50 mm yr⁻¹ under eucalyptus plantations to more than 400 mm yr⁻¹ in Cerrado pastures, where well-drained soils deeper than 1 m favor infiltration.

In Argentina, studies focus on a humid temperate climate (Pampas), where annual rainfall is approximately 1,000 mm yr⁻¹ and dominant land uses include soybean, wheat, and pastures. Reported recharge rates range from 150 to 400 mm yr⁻¹ (Varni et al. 2013). In contrast, semi-arid regions dominated by irrigated vineyards show considerably lower recharge, typically 50–150 mm yr⁻¹ under rainfall of 150 to 350 mm yr⁻¹, reflecting the intensive water use and limited precipitation (Jobbágy et al. 2011). In Central Chile, under a Mediterranean climate, the replacement of native vegetation by commercial pine plantations and irrigated fruit-growing systems has resulted in recharge values between 100 and 300 mm yr⁻¹, with reductions up to 30% relative to native cover (Alvarez-Garreton et al. 2019).

In other Latin American countries, such as Mexico, Colombia, and Peru, hydrological setting differs substantially. In Yucatán (Mexico), karst aquifers in a humid

tropical climate exhibit relatively high recharge rates, ranging from 200 to 600 mm yr⁻¹ (Bauer-Gottwein et al. 2011). In the Colombian Andes, under a tropical highland climate, coffee agroforestry systems help maintain high recharge rates of 300–700 mm yr⁻¹ (Buytaert et al. 2006).

In China, scientific production has been strongly focused on the impacts of perennial crops on aquifer recharge, with emphasis on the Loess Plateau, a region characterized by seasonal rainfall and semi-arid conditions (Li et al. 2019, 2023; Tao et al. 2021; Zhang et al. 2018, 2020; Huang et al. 2013, 2019; Jia et al. 2017; Wang et al. 2015). In this region, recharge rates often do not exceed 10 mm yr⁻¹, reflecting severe climate constraints and high-water consumption by agricultural systems. Perennial crops such as apple and peach orchards (Zhang et al. 2018, 2020; Li et al. 2019) have been associated with progressive depletion of groundwater reserves deeper than 20 m, with recovery that is slow and limited. This effect results from the combination of deep roots that extract water from great depths, low natural recharge under semi-arid conditions, intensive management practices, and soil properties that facilitate groundwater access but not rapid replenishment.

Isotopic studies conducted on the Loess Plateau showed that the isotopic signature of xylem coincides with that of water derived from deep soil layers exceeding 10 m depth (Huang et al. 2019; Tao et al. 2021). In apple orchards, up to 80% of the water uptake originated from depths greater than 5 m (Li et al. 2019), and during summer, deep roots supplied up to 60% of total water demand (Tao et al. 2021). Intensive use of this deep soil water reservoir has led to the desiccation of subsurface layers (Zhang et al. 2018), as identified through $\delta^{18}\text{O}$ and chloride profiles. Although the deep porous soil allows roots to penetrate easily and access water at great depths, low water-holding capacity and high evapotranspiration make the system vulnerable to long-term depletion (Li et al. 2019; Shi et al. 2021).

In North America, comparative studies show that pasture areas can exhibit recharge rates up to ten times higher than those recorded under planted forests, indicating that vegetation type influences not only infiltration volume, but also water transit time and the depth of percolation within the soil-aquifer system (Adane et al. 2018; Befus et al. 2017; Hubbard et al. 2010; Scanlon et al. 2007). Similar dynamics have been reported for woody species in the Brazilian Cerrado, which access water from shallow subsurface layers during the rainy season but rely on deep reserves (> 10 m) during the dry season (Oliveira et al. 2005a, b). The slow recovery of soil water storage, even after crop removal, further highlights the persistence of these impacts. These patterns reinforce the role of vegetation in modulating both through infiltration processes and through root water uptake.

Overall, land-use change alters recharge primarily through combined effects on evapotranspiration, soil structure, and infiltration pathways.

Influence of soil type and climatic conditions

While vegetation and land use strongly affect evapotranspiration and recharge, these responses are also conditioned by soil properties and precipitation regime. In addition to land use and vegetation cover, soil type and rainfall patterns play a fundamental role in groundwater recharge dynamics. In semi-arid regions, where water availability is naturally limited, constraints on recharge become particularly evident. On the Loess Plateau (China), for example, dominant land uses include grain crops (soybean, corn, and wheat), apple orchards, and pastures, yet recharge represents less than 5% of the mean annual rainfall of 532 mm (Li et al. 2019, 2023; Tao et al. 2021; Zhang et al. 2018, 2020; Huang et al. 2019), making groundwater replenishment difficult.

In contrast, regions with sandier soils, such as the Sand Hills in the United States (Adane et al. 2018), show substantially higher infiltration rates (97 mm yr⁻¹), especially under pastures, where low water demand favors recharge even at mean annual rainfall of 570 mm yr⁻¹. Thus, climatic regime and soil properties strongly control whether vegetation effects translate into effective recharge.

Methodological challenges and gaps

The reviewed literature highlights important methodological limitations in recharge estimation (Table 1). Tracer-based techniques (Maeght et al. 2013; Alaoui et al. 2011; Si 2007), for instance, may underestimate preferential flow pathways in soil. Widely used models, such as HYDRUS (Adane et al. 2018; Min et al. 2015), also present limitations when applied to tropical environments, as they are highly sensitive to parameters that are difficult to quantify, including unsaturated hydraulic conductivity, pore-size distribution, and effective rooting depth.

These technical limitations are compounded by gaps observed in the reviewed studies themselves, such as the scarcity of long-term datasets (longer than 20 years) needed to evaluate recharge trends amid land-use and land-cover changes. In addition, few studies provide detailed data on the effective rooting depth, a key variable for understanding water uptake and redistribution processes within the soil profile. There is also limited quantitative evidence on the impacts of irrigated agriculture on shallow aquifers, especially in regions where intensive irrigation may lead to localized groundwater overexploitation.

Table 1 Recharge estimation methods

Type	Method	Frequency (%)	Typical Application	Advantage	Limitation
<u>Direct</u>	Water table fluctuation (WTF)	17	Areas with continuous groundwater level monitoring	High accuracy at local scale	Requires dense monitoring network and long time series
	Lysimeters and in situ measurements	4	Controlled field experiments and plot-scale studies	High measurement accuracy	High cost and limited spatial representativeness
	Environmental tracers (Cl^- , ^3H , ^2H , ^{18}O)	11	Deep soils and systems with slow flow dynamics	Estimation of water age and origin	High analytical complexity and cost
<u>Indirect</u>	Water balance (P–ET)	21	Catchment-scale and regional assessment	Conceptual simplicity	High uncertainty in ET estimation
	Numerical Modeling (e.g. HYDRUS, InVEST)	27	Scenario analysis of land use and climate change	Simulation of future scenarios and system dynamics	High data and calibration requirements
	Baseflow separation	9	Streamflow time-series analysis	Low operational cost	Limited applicability in data-scarce regions
	Remote sensing	11	Large-scale spatial analysis using satellite data	Wide spatial coverage	Limited temporal and spatial resolution

Method selection is strongly influenced by study scale, research objectives, and data availability. In general, direct methods are applied in site-specific or validation studies, whereas indirect methods predominate in regional analyses and predictive modeling. When feasible, combining multiple approaches is an effective strategy to reduce uncertainty and improve the robustness of recharge estimates.

Method sensitivity also varies according to land use (Table 2). In forests and orchards, direct methods report recharge values 30–50% lower than those obtained from modeling in arid climates, reflecting their ability to capture

water uptake from deeper reserves. In contrast, in pastures and agricultural areas, this difference is smaller (<20%), consistent with more homogeneous and superficial recharge processes.

Values compiled from available comparable studies; sample size varied by category.

Extreme cases illustrate these methodological discrepancies. In Brazil, for eucalyptus plantations in humid climates, water table fluctuation (WTF) recharge estimates (318 mm yr^{-1}) were 38% lower than those from modeling (514 mm yr^{-1}). In contrast, studies of orchards in northern China reported zero recharge using isotopic methods, whereas a simple water balance estimated 25 mm yr^{-1} , suggesting a positive bias in indirect estimates. This difference may be partly related to the temporal resolution and detection depth of each method. Water balance approaches tend to reflect short-term surface infiltration, including water stored in the vadose zone that does not necessarily reach the aquifer, especially in thick soils or under deep-rooted vegetation. In contrast, isotopic methods capture effective recharge that penetrates beyond this zone and contributes to groundwater flow, typically integrating long-term processes. These methodological differences help explain the contrasting results among approaches.

In semi-arid climates, indirect methods tend to overestimate recharge relative to direct methods by approximately 10% to 80%, depending on the methodology, with discrepancies up to twice that amount. In humid climates, this difference appears smaller, with variations of about 15% to 30% between methods.

When the analysis is segmented by vegetation type, consistent patterns emerge in methodological discrepancies. In planted forests, direct methods report recharge values 40% to 60% lower than those obtained from indirect approaches, whereas in pastures, the opposite pattern is observed, with direct methods indicating values 10% to 30% higher. These discrepancies are primarily associated with limitations of each method in capturing key ecohydrological processes.

In semi-arid regions, hydrological models often fail to adequately represent water uptake by deep roots, leading to overestimation of recharge by indirect methods. In tropical areas, water balance approaches may also overestimate recharge in clay-rich soils, particularly when water retention and lateral flow processes are not considered.

Table 2 Range of groundwater recharge values (mm yr^{-1}) by estimation method, land use type and climatic condition, compiled from reviewed studies

Type	Method	Planted Forests		Orchards		Pastures		Agriculture	
		Arid	Wet	Arid	Wet	Arid	Wet	Arid	Wet
Direct	Monitoring (WTF)	5–50	50–150	0–15	10–40	20–100	100–300	30–120	150–400
	Isotopes	2–30	40–120	0–10	5–30	15–80	90–250	25–100	130–350
Indirect	Modelling	10–60	70–180	5–20	20–60	30–120	150–350	50–150	200–450
	Water Balance	15–70	80–200	0–25	30–80	40–150	180–400	60–180	250–500

These differences may be associated with limitations such as:

- Only 28% of the reviewed studies applied both direct and indirect methods within the same study area, limiting the systematic comparison of divergences.
- Approximately 80% of the data derived from direct methods are based on monitoring periods longer than five years, which complicates comparison with annual or short-term estimates obtained from indirect approaches.

Additional factors such as rooting depth, temporal scale of observation, and methodological spatial resolution (Table 3) may contribute to discrepancies among results and, in some cases, may be interpreted as systematic biases. Indirect methods may underestimate groundwater consumption by deep-rooted vegetation and smooth out local variations in heterogeneous landscapes.

The analysis also revealed evidence of methodological bias:

- 68% of the studies reporting recharge values greater than 100 mm yr^{-1} used modeling approaches;
- 73% of the studies reporting recharge values lower than 10 mm yr^{-1} were based on direct methods.

Moreover, the reviewed studies show limited standardization in uncertainty reporting, with confidence intervals ranging from approximately $\pm 5\%$ (isotopic methods) to $\pm 35\%$ (simple water balance approaches). Therefore, recharge estimates should be interpreted with caution, considering methodological assumptions as well as spatial and temporal scale.

Deep aquifer recharge

The results presented in Sect. 3.1 to 3.3 represent a structured synthesis of the selected literature, in which consistent patterns, methodological contrasts, and ecohydrological mechanisms were identified and compared across studies, climates, soil types and land use categories.

Table 3 Factors influencing the variation between recharge estimation methods

Factor	Impact on Results	Observed
Root depth	Underestimation by indirect methods	WTF detects near-zero recharge; model estimates $15\text{--}30 \text{ mm}$ (e.g., orchards)
Temporality (time scale)	WTF captures extreme events; models represent averages	WTF captures short-term extremes (e.g., -5 mm yr^{-1} during drought); model tend to remain positive
Spatial resolution	Spatial methods smooth heterogeneities	Fragmented pastures: $\pm 20\%$ error

Because the reviewed studies employed different recharge estimation methods, reported values are not always directly comparable. Direct methods more often reflect recharge reaching the aquifer, whereas indirect methods may also represent modeled or residual percolation depending on underlying assumptions. Therefore, the numerical ranges presented here should be interpreted as comparative indicators rather than strictly uniform quantities.

(I) land use and land cover changes and their effects on recharge

Deep-rooted vegetation ($> 5 \text{ m}$), characteristic of natural formations such as tropical forests and Cerrado biome, plays an essential role in aquifer recharge by enhancing infiltration, reducing surface runoff, and increasing soil water retention (Shi et al. 2021; Turkeltaub and Bel 2024). However, conversion of these areas to agricultural uses, especially intensive monocultures and pastures, can lead to soil compaction and reduced infiltration capacity, thereby increasing runoff, intensifying erosion, and decreasing effective recharge (Scanlon et al. 2007). Importantly, the effect of deep-rooted vegetation on recharge depends not only on root depth but also on vegetation type and soil management. While native systems generally favor infiltration and recharge maintenance, anthropogenic uses such as homogeneous eucalyptus plantations or intensive agriculture may significantly reduce recharge due to high water demand, rainfall interception by the canopy, reduced turnover, and discontinuous vegetation cover (Lucas and Wendland 2016; Mattos et al. 2019).

Intensive agricultural systems are also associated with high evapotranspiration and surface runoff rates, partly because soil remains exposed between crop cycles, unlike natural vegetation or perennial systems that maintain continuous cover throughout the year (Lucas and Wendland 2016). Owuor et al. (2016) and Scanlon et al. (2007) suggest that converting natural vegetation to agriculture can increase surface runoff by 5–20% and reduce evapotranspiration by 10–30%, depending on crop type and climate. In addition, exposed soils between crop cycles may increase surface runoff to approximately 15–40% of precipitation, compared with 5–20% under continuous vegetation cover.

These impacts can be quantified using vegetation indices such as the Leaf Area Index (LAI) and the Normalized Difference Vegetation Index (NDVI), which are widely used to monitor vegetation dynamics and land use effects on the water balance (Júnior et al. 2022). Studies indicate that agricultural areas in the Cerrado exhibit lower NDVI values during the dry season, reflecting temporary loss of vegetation cover and increased surface runoff (Trentin et al. 2021). In contrast, agroforestry systems and native vegetation tend to maintain higher NDVI and LAI values throughout the year, which are

Table 4 Impact of vegetation types on water recharge

Type of Vegetation	Recharge (%P)	R/P (mm yr ⁻¹)	Observed Impact
Natural pastures	15–25	150–250	High recharge (low water extraction), except in compacted soils
Planted forests (e.g., eucalyptus, pine)	5–30	50–300	Significant recharge reduction (30–60%) in compared to pasture due to high ET and deep root water uptake
Orchards (e.g., apple, peach)	2–10	20–100	Deep water uptake; potential depletion of soil moisture and groundwater
Annual agriculture (e.g., sugarcane, soybeans)	10–20	100–200	Variable, depending on irrigation and management
Native vegetation (Cerrado)	5–12	50–120	High infiltration capacity; recharge depends on soil structure and vegetation competition

Values compiled from available comparable studies; sample size varied by category

Table 5 Average ET/P ratio by type of land use

Type of Land Use	Mean ET/P (%)	ET/P (mm yr ⁻¹)	Observed Interval (%)
Planted forests (eucalyptus)	85	850	70–95
Orchards (apple, peach)	75	750	60–90
Native forests	78	780	65–92
Natural pastures	65	650	50–82
Annual crops	70	700	55–85
Savannahs/Cerrado	82	820	75–90

Values compiled from available comparable studies; sample size varied by category

associated with greater rainfall interception and infiltration capacity (Wang et al. 2015; Varnier et al. 2023), although not necessarily with higher groundwater recharge rates.

The methods used to estimate groundwater recharge in the selected studies reveal a predominance of indirect approaches (70%), with emphasis on numerical modeling (27%), which is widely used for its ability to simulate land-use and climate-change scenarios, although it requires detailed input data and careful calibration. Water balance approaches were also commonly applied at regional scales (21%) due to their conceptual simplicity, but they are subject to uncertainties, particularly in evapotranspiration estimates. Other indirect methods included baseflow separation (9%), applied where long-term streamflow records are available, and remote sensing (11%), which is useful for large-scale analyses but requires field validation.

Among direct methods (30%), the water table fluctuation (WTF) method was the most frequently used (17%) and applied in areas with consistent groundwater-level time series. In contrast, lysimeters (4%) and isotopic or

Table 6 Factors influencing the ET/P ratio and their typical effects based on reviewed studies

Factor	Effect on ET/P	Example
Vegetation age	Increase of 0.5–1.5% per year	20-year-old orchards show 15–20% higher ET/P (Zhang et al. 2020)
Root depth	Increase of up to 30%	Species with roots deeper than 5 m (Li et al. 2019)
Soil texture	Increase of 10–15% in sandy soils	Higher ET losses compared to clay soils (Huang et al. 2018)
Water availability	Reducing of up to 30%	Seasonal variation in water availability

Values compiled from available comparable studies; sample size varied by category

chemical tracers (11%) were less commonly applied due to their higher cost and analytical complexity, despite offering greater precision at local scale.

Vegetation plays a key role in modulating recharge across land use types. The compiled evidence is summarized in Table 4, which presents recharge (R) in relation to precipitation (P).

The ratio of evapotranspiration (ET) and precipitation (P), expressed as ET/P (%), varied substantially among land-use categories across the reviewed studies (Table 5). The highest mean ET/P values were observed in planted forests (85%) and savannah/Cerrado systems (82%), whereas pastures showed the lowest mean value (65%).

Annual crops showed intermediate ET/P values (70%), reflecting shorter growing cycles, shallower root systems, and periods of bare soil. Pastures exhibited greater variability (50–82%), associated with differences in species composition, vegetation cover, and grazing management. Extreme values have been recorded in mature orchards on the Loess Plateau (ET/P=109–114%) and in Eucalyptus plantations during dry years in Brazil (ET/P up to 120%), typically interpreted as evidence of water deficit and use of stored soil moisture.

Biophysical factors strongly modulate the ET/P ratio (Table 6). Root depth and vegetation age were identified as key controls, with ET/P increasing as root system deepen (>5 m) or as vegetation matures. Soil texture also plays an important role, with sandy soils often associated with higher ET/P values. In contrast, water availability is a limiting factor, particularly during dry periods, leading to reduced ET/P ratios in some land-use categories.

An inverse relationship between ET/P and recharge was consistently observed:

- ET/P < 60% associated with high recharge (> 30% of P).
- ET/P > 80% indicated low recharge (< 10%).
- ET/P > 100% correspond to negligible or negative recharge, implying direct use of water stored in the soil.

Table 7 Ranges of ET/P ratio (%) by land use in different climates

Climate	Forests (%)	Orchards (%)	Pastures (%)
Arid	90–95	80–90	70–85
Temperate	75–90	65–80	55–75
Humid tropical	70–85	60–75	50–70

Values compiled from available comparable studies; sample size varied by category

A representative example was reported for an orchard with $ET/P = 105\%$, in which no recharge occurred and the resulting water deficit (5% of precipitation) was compensated by subsurface water reserves (Huang et al. 2018).

Analysis by climatic zone revealed systematic variations. In semi-arid regions, the ET/P ratio was particularly high, especially in forests (92%) and orchards (98%), contributing to the frequent occurrences of negligible or negative recharge (Table 7).

The reviewed studies indicate that conversion of agricultural land to pasture can significantly alter local hydrological processes, especially infiltration and recharge. Although pastures exhibit lower Leaf Area Index (LAI) and therefore lower evapotranspiration than crops or forests, their vegetation is dominated by shallow root systems (<1 m), which limit water percolation into deeper soil layers. In addition, soil structural degradation associated with pastures conversion, often intensified by cattle trampling, reduces porosity and water retention capacity, with reported increase in soil bulk density of up to 3% and reductions in infiltration of about 57% in infiltration, thereby enhancing surface runoff (Tuffour et al. 2014; Drewry et al. 2008; Mapfumo et al. 2000). As a result, aquifer recharge may decline even when surface infiltration increases.

Lower evapotranspiration in pastures does not necessarily translate into higher effective recharge. Recharge efficiency depends not only on the precipitation-ET balance but also on vertical soil connectivity. This connectivity is enhanced by deep-rooted vegetation, which creates conductive macropores, whereas compacted soil under pastures

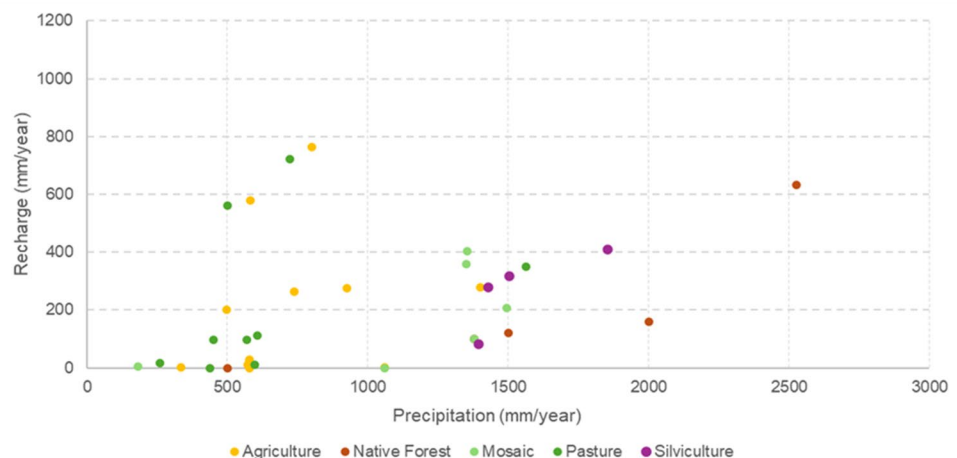
restricts vertical flow, favoring water losses through surface runoff and evaporation.

Similarly, intensive agricultural systems are associated with higher evapotranspiration rates, partly due to reduced vegetation cover and increased soil exposure between crop cycles. Although reduced canopy interception in pastures allows more rainfall to reach the soil surface compared with forests, surface compaction limits infiltration and percolation. Consequently, a substantial fraction of the infiltrated water remains in near-surface layers, where it is lost through evaporation or uptake by shallow roots, rather than contributing to groundwater recharge.

In contrast, forested systems exhibit distinct hydrological behavior. Despite higher rainfall interception and evapotranspiration, deep-rooted vegetation promotes the formation of macropores that act as preferential flow pathways, rapidly conducting water beyond the evaporation zone and shallow root layers, thereby increasing recharge efficiency. Thus, effective recharge depends critically on vertical soil connectivity provided, which is enhanced by deep-rooted vegetation and reduced in compacted soils. Conversely, the absence of continuous vegetation cover limits infiltration and promotes rapid evaporation of soil moisture, reducing the amount of water available for recharge.

Figure 4 summarizes recharge patterns across land uses. Native forests and planted forests often exhibit high recharge in humid climates with well-structured, deep soils, whereas agricultural systems and pastures are more variable and strongly influenced by management practices and soil compaction. In semi-arid climates, recharge remains low across all land use types.

Among grouped land uses, native forested areas showed the highest mean recharge, whereas planted forests exhibited high but highly variable values. Forest areas showed an average recharge of 405 mm yr⁻¹ under mean annual precipitation of 1575 mm and evapotranspiration (ET) of 912 mm yr⁻¹. Despite high ET losses, high root density, continuous

Fig. 4 Recharge for different land uses and occupations

soil cover, and hydraulic redistribution favor deep infiltration and contribute significantly to aquifer recharge. In native tropical forests, recharge values exceeding 400 mm yr⁻¹ were reported even with ET above 950 mm yr⁻¹, reflecting the high hydrological efficiency of this land cover. Hydraulic redistribution has been identified as an important subsurface water flux during dry periods, supporting vegetation activity and influencing both atmospheric fluxes and subsurface storage. Under such conditions, deep-rooted vegetation may simultaneously enhance both evapotranspiration and infiltration, resulting in substantial recharge even at high ET rates.

Pasture areas showed a mean recharge of 370 mm yr⁻¹, with intermediate annual precipitation (895 mm yr⁻¹) and high ET (824 mm yr⁻¹). Low recharge values, despite substantial rainfall, are attributed to soil compaction, reduced water retention capacity, limited vegetation cover, and the predominance of shallow root systems, which restrict infiltration and favor surface runoff. In semi-arid to arid regions, recharge remains limited, even under moderate ET (average of 895 mm yr⁻¹), not exceeding 100 mm yr⁻¹. This condition is associated with shallow herbaceous vegetation and degraded soil structure, which reduce infiltration capacity and increase runoff.

In agricultural areas, mean recharge was 218 mm yr⁻¹, with substantial variability driven by climate, crop type, and management practices. Despite relatively low rainfall in some regions (around 690 mm yr⁻¹), recharge values exceeding 400 mm yr⁻¹ have been reported in systems with well-drained soils and management practices that promote infiltration. In semi-arid to arid climates, mean ET reached 929 mm yr⁻¹ with mean recharge of 297 mm yr⁻¹, whereas in semi-humid subtropical regions, ET increased to 1252 mm yr⁻¹ while recharge declined to 106 mm yr⁻¹, indicating reduced hydrological efficiency under high evaporative demand. In humid regions, higher ET values are associated with greater water availability and vegetation cover, while recharge may be constrained by shallow, compacted, or saturated soils rather than precipitation.

Areas characterized by mixed or mosaic land use exhibited intermediate hydrological behavior, with a mean recharge of 216 mm yr⁻¹ and ET ranging from 900 to 1300 mm yr⁻¹, reflecting heterogeneous vegetation cover and soil conditions. In a semi-humid subtropical climate, recharge value of 324 mm yr⁻¹ was observed under mean ET of 995 mm yr⁻¹, suggesting that landscape heterogeneity and the presence of permeable zones can enhance infiltration even under high evaporative demand.

Planted forests, particularly in tropical and subtropical regions, exhibit high ET (894–1394 mm yr⁻¹) and a mean recharge of 280 mm yr⁻¹, with substantial variability among studies. Eucalyptus plantations tend to show lower recharge

(82–120 mm yr⁻¹), whereas systems with mixed vegetation or greater effective soil depth can exceed 400 mm yr⁻¹. This pattern reflects strong, deep-root water uptake combined with potential limitation to infiltration due to canopy interception, reduced understory cover, and changes in soil structure.

Overall, the dynamics between evapotranspiration and recharge is non-linear and controlled by multiple factors. Variables often omitted from conventional soil water-balance approaches, such as root system architecture and vertical flow processes in deep-rooted vegetation, significantly affect the partitioning between ET and recharge. As a result, studies in forested and planted forest systems with high ET frequently report lower recharge when estimates are derived solely from simplified water balance methods.

Analysis of recharge data by climatic zone reveals clear patterns in the hydrological efficiency of different land uses across environmental contexts. In humid tropical climates, the highest mean recharge values were associated with forested areas and planted forests (Alvarez-Garreton et al. 2019; Mattos et al. 2019; Adane et al. 2018), where permanent vegetation cover and deep root systems favor percolation despite high evapotranspiration rates.

In subtropical climates, mosaic systems exhibited favorable hydrological performance, combining native vegetation (with recharge rates approximately 30–50% of precipitation) and anthropogenic land uses, thereby maintaining relatively high recharge, especially in regions with well-distributed seasonal rainfall (Oliveira et al. 2019; Lucas and Wendland 2016; Wendland et al. 2007).

In contrast, semi-arid regions consistently showed low recharge values regardless of land use (Zhang et al. 2018; Owuor et al. 2016; Li et al. 2019), reflecting limited rainfall and, in many cases, degraded soil structure that restricts infiltration. Also, according to Owuor et al. (2016) and Zhang et al. (2018), converting pastures to forests may significantly reduce recharge in semi-arid regions, with stronger effects under more arid conditions.

Overall, the results indicate that the land use most favorable to recharge depends on the interaction among vegetation cover, soil properties, and climatic regime. Higher recharge values are generally associated with forested systems in humid tropical climates, whereas the lowest values are typically observed in pastures under semi-arid conditions.

A more detailed analysis of recharge across vegetation types and climatic contexts reveals consistent patterns in hydrological efficiency of land uses. In general, planted forests, including eucalyptus, pine, and orchards, tend to exhibit the lowest annual recharge values across the evaluated climates. In semi-arid regions, recharge under these systems is extremely limited, ranging from 0.7 to 17 mm yr⁻¹, which corresponds to only 1–5% of annual rainfall (Zhang et al. 2018; Adane et al. 2018). In humid tropical

climates, recharge values under planted forests remain low to moderate, ranging from 0.7 to 96.5 mm yr⁻¹ (Mattos et al. 2019). This pattern is associated with high evapotranspiration rates and deep root systems capable of accessing groundwater reserves.

Pastures and native vegetation exhibit comparatively high recharge when evaluated solely in terms of recharge-to-precipitation ratios. In semi-arid regions, recharge under these covers may reach up to 42% of precipitation (Owuor et al. 2016), while in subtropical climates reported values range from 47% to 51% (Oliveira et al. 2019). In tropical regions, recharge under pastures ranges from 300 to 407 mm yr⁻¹ (Wendland et al. 2007), corresponding to approximately 30% of precipitation. However, when evapotranspiration is considered, it may account for up to 92% of precipitation, which substantially reduces recharge efficiency. Annual cropland systems show considerable variability across climatic zones. In semi-arid regions, recharge ranges from 50 to 55 mm yr⁻¹ (Zhang et al. 2018), whereas in subtropical climates, recharge is approximately 15–24% of precipitation, or about 150–200 mm yr⁻¹ (Lucas and Wendland 2016). In tropical climates, recharge values of approximately 200 mm yr⁻¹ have been reported (Min et al. 2015). This variability is influenced by management practices, including irrigation, tillage, and mulching.

A synthesis of these results is presented in Table 8, which summarizes recharge as a proportion of precipitation across climate zones and land use types. These patterns represent generalized trends derived from the reviewed studies and may vary with soil properties, vegetation characteristics, and estimation methods.

Across the reviewed studies, pastures and native vegetation were frequently associated with higher recharge proportions, especially in semi-arid climates, where water limitation increases the sensitivity of infiltration processes to land use. In contrast, planted forests, such as eucalyptus and pine plantations, exhibited lower recharge values, particularly under low rainfall conditions. Annual cropland systems showed intermediate behavior, with recharge outcomes strongly influenced by soil properties, management intensity, and rainfall regime.

Table 8 Groundwater recharge tendencies (as % of precipitation) by climate context

Climate	Recharge ranking (highest to lowest)
Arid	Pasture (15–25%) > Agriculture (10–20%) > Mosaic of Uses (8–15%) > Native Forest (5–12%) > Forestry (2–10%)
Subtropical	Pasture (25–30%) > Native Forest (20–25%) > Mosaic of Uses (15–20%) > Agriculture (10–15%) > Forestry (5–15%)
Tropical	Pasture (20–25%) > Native Forest (18–22%) > Mosaic of Uses (12–18%) > Agriculture (8–12%) > Forestry (5–10%)

Notably, although pastures may exhibit relatively high recharge-to-precipitation ratios and lower evapotranspiration-to-precipitation (ET/P) values, this does not necessarily translate into effective aquifer recharge. These estimates often do not account for factors such as soil compaction, reduced porosity, shallow root systems, and the loss of macropores and preferential vertical pathways. Under such conditions, apparent recharge may reflect only temporary increases in soil moisture within the topsoil and the shallow subsurface, which are highly susceptible to evaporation. Thus, a significant portion of infiltrated water may be rapidly returned to the atmosphere rather than contributing to groundwater recharge.

Conversely, native forests, although exhibiting higher evapotranspiration rates and lower apparent recharge than pastures, tend to promote deeper, more persistent infiltration. Extensive root systems create macropores and preferential flow pathways that facilitate the movement of water beyond the surface zone, thereby contributing to effective aquifer recharge. This process is, is often not properly captured by conventional water balance approaches.

According to Kabir (2005), inadequate management practices, such as excessive and repeated application of agrochemicals and the absence of conservation practices, are associated with soil structural degradation. Evidence indicates that intensive use of pesticides, herbicides, and synthetic fertilizers can negatively affect flora and fauna by suppressing key organisms, including mycorrhizal fungi, bacteria, and soil invertebrates. The decline in biological activity compromises aggregate formation and stabilization, leading to reduce macroporosity and loss of granular structure, both essential for water infiltration and moisture storage within the soil profile. These effects, combined with soil exposure and traffic from machinery or livestock, increase surface compaction and susceptibility to both sheet and rill erosion.

However, conservation practices have been identified as effective strategies for mitigating the negative impacts of land-use change on aquifer recharge (Min et al. 2015). Overall, land use alone does not determine recharge; outcomes emerge from interactions among vegetation type, management practices, soil structure, and climate.

(II) influence of deep roots on evapotranspiration and recharge

Across the reviewed studies in biomes such as the Brazilian Cerrado (Oliveira et al. 2005a, b), the African savannas (Chung et al. 2016; Xu and Beckman 2019), and tropical forests of Indonesia (Zhang et al. 2020), root system depth consistently emerges as a key factor controlling evapotranspiration patterns and groundwater recharge. In these environments, deep-rooted species can access water reserves at depths exceeding 10 m, particularly during prolonged dry

periods. This sustained groundwater uptake allows vegetation to maintain physiological activity even under surface water deficit (Bleby et al. 2010; Fan et al. 2017).

The literature further indicates that water uptake by deep roots may involve groundwater stored over multiple years or even decades, implying that a portion of transpired water is not derived from recent precipitation (Zhang et al. 2017; Miguez-Macho and Fan 2021). This process challenges conventional hydrological models, which often assume that evapotranspiration originates solely from recent rainfall stored in shallow soil layers. When deep water uptake is not explicitly represented, such models may overestimate aquifer recharge by attributing all percolating water to recharge, without accounting for subsequent extraction by roots (Li et al. 2023). Consequently, part of the infiltrated water may be misinterpreted as effective recharge, even though it is later removed by vegetation, highlighting a key limitation in current recharge assessments.

However, this simplified perspective can lead to misinterpretations regarding the role of forests and large trees in aquifer recharge. The widespread assumption that “forests reduce aquifer recharge” is largely based on their high evapotranspiration rates, but overlooks key ecohydrological mechanisms such as root-induced macropore formation, hydraulic redistribution, and preferential flow pathways, which may enhance effective recharge under certain conditions (Bargués Tobella et al. 2014; Ilstedt et al. 2007; Jiang et al. 2021).

Macropores formed by roots, soil fauna, or structural features act as low-resistance pathways that accelerate water infiltration, reduce surface runoff, and may increase recharge by up to 60% in forest soils (Alaoui et al. 2011). In areas transitioning from pasture to forest, increases in infiltration of 15–40% have been associated with enhanced bioturbation and improved soil structure (Min et al. 2015).

The reviewed studies collectively indicate that vegetation effects on recharge are not one-dimensional but arise from the interaction among root water uptake, preferential infiltration processes, and soil water storage capacity. In shallow soils or in systems dominated by shallow root systems, recharge may be overestimated when models assume that all infiltrated water contributes to aquifer recharge, without accounting for retention in the soil profile or subsequent uptake by vegetation.

Although macropores can accelerate infiltration, their effect is reduced in sandy soils, where matrix flow is already efficient. In these contexts, recharge can be strongly limited by deep root water uptake, as noted by Li et al. (2019) and Zhang et al. (2018). Similarly, Min et al. (2015) observed a 25% increase in recharge in non-compacted agricultural soils associated with higher microporosity, whereas in sandy soils the effect was typically less than 10%. Moreover, deep-rooted vegetation may reduce recharge through

hydraulic uptake, where roots extending beyond 5 m access water below the active recharge zone, increasing evapotranspiration and limiting deep percolation. In the Loess Plateau (China), apple orchards with roots exceeding 10 m showed near-zero annual recharge (Li et al. 2019), while in Brazil, eucalyptus plantations reduced recharge by 30–80% (Zhang et al. 2018).

Another mechanism frequently associated with deep-rooted vegetation is hydraulic redistribution (HR), which indirectly helps maintain soil moisture and enhance subsequent infiltration (Belovitch et al. 2022). HR is a passive process of water movement through plant root systems, driven by gradients in soil water potential. During the night, when transpiration is minimal and the stomata are closed, roots act as conduits, transferring water from wetter, deeper soil layers to drier, shallower layers (Oliveira et al. 2005a, b; Prieto et al. 2012; Belovitch et al. 2022). This upward flow occurs without metabolic energy expenditure and is governed solely by differences in soil moisture and water potential.

This process temporarily increases soil moisture in near-surface layers, benefiting shallow-rooted plants, sustaining microbial activity, and facilitating the infiltration of subsequent rainfall events. In addition, preferential flow pathways associated with roots and root-derived macropores enhance vertical water movement (Bargués Tobella et al. 2014; Ilstedt et al. 2007). These macropores act as highly conductive channels that accelerate percolation beyond the soil matrix, particularly in well-structured tropical soils.

Thus, deep-rooted vegetation may promote more efficient recharge than agricultural systems or pastures (Belovitch et al. 2022; Jiang et al. 2021), under conditions of high soil porosity, continuous macropore networks, medium to sandy texture, and well-drained profiles, combined with seasonal rainfall regimes typical of humid tropical climates. However, this effect is not universal. Huang et al. (2018) demonstrated that even a 20% increase in infiltration due to macropores can reduce annual recharge by up to 80%, highlighting that the net effect depends on the balance between competing processes. The interaction between macropore development and deep root water uptake is strongly controlled by soil properties, as summarized in Fig. 5.

The magnitude and effectiveness of hydraulic redistribution depend on factors such as root system depth and density, soil moisture gradients, soil hydraulic conductivity, plant maturity stage, and vegetation type. While dense vegetation can increase evapotranspiration due to the greater leaf area, it may also enhance deep infiltration by improving soil structure, reducing surface runoff, and increasing vertical connectivity through root-induced macropores (Belovitch et al. 2022; Bargués Tobella et al. 2014; Prieto et al. 2012). However, this balance is context dependent. Li et al. (2023) suggest that hydraulic redistribution may reduce aquifer

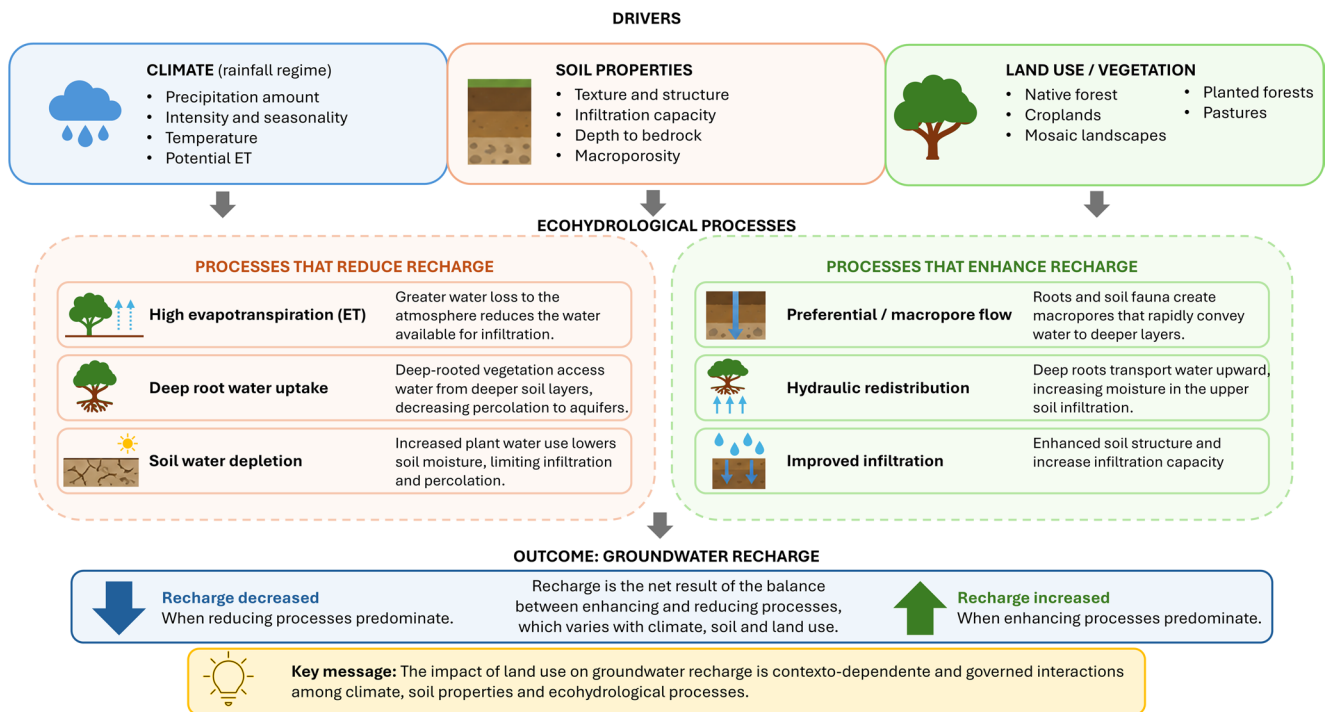


Fig. 5 Context-dependent effects of forests and deep roots on aquifer recharge

recharge in areas where root water uptake exceeds rainfall replenishment. In Brazil, Zhang et al. (2018) reported that eucalyptus plantations reduce recharge by up to 80%. Such species can access stored water below the active recharge zone, thereby reducing deep percolation. In humid climates, this impact persists, but is typically less pronounced, with reductions of approximately 20–40%.

The effective depth of root system varies according to soil texture and profile thickness. Deep, sandy soils favor extensive root development, with depths exceeding 18 m (Zhang et al. 2018), while shallow or clay-rich soils tend to restrict root growth to less than 40 cm (Costa 2012; Coelho et al. 2001; Ceballos et al. 2002).

In ecosystems such as the Brazilian Cerrado and the Amazon rainforest (Scholz et al. 2002; Oliveira et al. 2005a, b), some tree species extract water from deep soil layers during the dry season and subsequently redistribute part of this water to shallower layers. This process benefits shallow-rooted plants by increasing near-surface moisture through hydraulic redistribution (Belovitch et al. 2022).

The positive effects of deep roots (> 5 m) on preferential flow (via macropores), enhanced infiltration, and intra-soil water redistribution suggest that, under specific conditions, particularly in well-structured soils and climates characterized by intense rainfall events interspersed with dry periods, dense tree vegetation may contribute significantly to aquifer recharge by increasing infiltration efficiency (Ilstedt et al. 2007; Bargués Tobella et al. 2014; Jiang et al. 2021).

However, the net effect of deep roots is context dependent: they may either reduce or enhance recharge depending on whether water uptake or preferential flow processes dominate under local soil and climatic conditions.

Discussion

Rooting depth is strongly influenced by land use, soil properties, and water availability, with several studies reporting roots exceeding beyond 20 m (Fan et al. 2017; Terada et al. 2022). In tropical forests, deeper root systems in tall trees have been associated with increased drought resistance (Molina et al. 2019; Zhang et al. 2018; Fan et al. 2017). Oliveira et al. (2005a, b) estimated that approximately 82% of evapotranspiration in forested environments originates from soil layers deeper than 5 m, which may lead to underestimation of both evapotranspiration and recharge in conventional soil water balance approaches. These findings suggest that, depending on climatic conditions and water-table depth, vegetation may access both long-residence-time groundwater and more recent rainfall-derived soil water, with direct implications for groundwater storage dynamics. Li et al. (2023) further reported that soil water storage declined within five years after the conversion of agricultural land or pasture to planted forests, with losses reaching up to 1400 mm yr⁻¹ in mature forests, in some cases exceeding mean annual precipitation.

A key limitation of the present review is the limited availability of studies that explicitly link vegetation controls, deep groundwater uptake, and their combined effects on evapotranspiration and recharge in the water balance.

One strand of the literature argues that continuous water extraction by deep-rooted vegetation can significantly reduce effective recharge and compromise groundwater storage over long time scales (Fan et al. 2017; Kleidon and Heimann 1998). Hydrological models that do not adequately consider upward water fluxes from deeper soil layers or groundwater tend to overestimate groundwater availability and underestimate evapotranspiration losses (Schymanski et al. 2008; Oliveira et al. 2005a, b). Moreover, by failing to distinguish between potential percolation within the soil profile and effective recharge reaching the aquifer, especially in vegetated soils with deep roots, such models reproduce one of the main conceptual limitations of conventional water balance approaches.

More recent studies, however, suggest that the negative effects of water extraction may be partially offset by ecohydrological mechanisms associated with deep-rooted vegetation, such as hydraulic redistribution and enhanced infiltration (Neumann and Cardon 2012; Prieto et al. 2012). Hydraulic redistribution increases residual soil moisture in drier soil layers, maintaining conditions closer to field capacity and favoring water percolation during rainfall events (Ryel et al. 2002; Oliveira et al. 2005a, b), thereby indirectly contributing to recharge in shallow aquifers. By failing to represent this mechanism, many models underestimate vegetation-infiltration interactions and may generate artificial estimates of evapotranspiration losses or residual recharge without process-based traceability.

The influence of vegetation on groundwater recharge is therefore conditioned by multiple factors: root density and depth (Fan et al. 2017), climatic characteristics (Bargués-Tobella et al. 2020), soil type (Ceballos et al. 2002; Coelho et al. 2001), precipitation regime, and land use/land cover (Lucas and Wendland 2016; Zhang et al. 2018).

Overall, the reviewed evidence here indicates that forests and deep-rooted vegetation do not exert a universally positive or negative effect on aquifer recharge. Instead, recharge outcomes depend on the balance between water losses through evapotranspiration and deep root uptake, and water gains promoted by preferential flow, root-derived macropores, hydraulic redistribution, and improved soil structure (Fig. 5). Recharge reduction tends to dominate in semi-arid regions, under low rainfall, deep soils, or fast-growing monoculture plantations with high water demand. In contrast, recharge maintenance or enhancement is more likely under humid tropical climates, seasonal rainfall regimes, and structured or cracked soils where preferential flow pathways are well developed. These findings reinforce the

idea that vegetation effects on recharge are strongly context-dependent and should be evaluated jointly with climate, soil hydraulic properties, and land management.

The development of macropores promoted by deep root growth increases vertical connectivity within the soil profile and facilitates preferential flow of water to deeper layers, thereby improving recharge efficiency (Pierret et al. 2016). These biological flow paths, formed by living or already-decomposing roots, represent low-resistance pathways for infiltration and tend to be more pronounced in well-developed forest systems. Under extremely dry soil conditions, however, rapid rainfall infiltration may trap air within soil pores and reduce hydraulic conductivity, limiting recharge. In such cases, hydraulic redistribution and the associated increase in soil moisture may enhance infiltration and partly mitigate this effect in drier soils (Kool et al. 2014). This hydraulic regulatory role of vegetation is rarely incorporated into conventional analyses, which often treat vegetation only as a water sink rather than as an active control on soil moisture dynamics.

Across the reviewed studies, the main factors associated with greater aquifer recharge include: (i) deep and well-structured soils; (ii) vegetation with deep and dense roots systems; (iii) continuous vegetation cover that reduces surface runoff and promotes infiltration; (iv) presence of macropores and high soil porosity; (v) hydraulic redistribution that maintains near-surface moisture; and (vi) low soil compaction. These elements, when present together, help maximize vertical water percolation in the profile and groundwater storage (Scanlon et al. 2007; Oliveira et al. 2005a, b; Pierret et al. 2016; Ryel et al. 2002). Recharge should not be estimated solely from climatic variables (such as net precipitation) but rather from an integrated understanding of these biophysical factors.

Studies addressing the impact of pastures on groundwater recharge report contrasting outcomes. Some studies indicate that converting agricultural land to pasture can promote soil degradation and reduce infiltration due to cattle trampling and surface compaction (Jia et al. 2017; Huang et al. 2013). Others suggest that well-managed pastures can maintain moderate to high infiltration rates, particularly when combined with sustainable management practices (Min et al. 2015; Huang et al. 2013; Alaoui et al. 2011), thereby allowing higher infiltration rates than unmanaged pastures. Factors such as stocking density, pasture rotation, and vegetation cover are therefore critical for preserving soil structure and maintaining water retention capacity.

Furthermore, few aquifer recharge studies provide evidence that groundwater flows beyond the physical limits of the hydrographic basin and drains below the surface outlet. This deep lateral flow occurs in response to regional hydraulic gradients and may represent a significant component of

the catchment water balance. Many models consider only surface flow as a basin outlet, overlooking that a significant part of the water budget may cross the basin boundaries through deep groundwater flow (Schymanski et al. 2008; Rempe and Dietrich 2018).

However, because it is not captured by conventional fluviometric stations, this groundwater flow component is often disregarded, leading to underestimation of the total volume of water that actually leaves the basin. Such omission can compromise the accuracy of water balance models and distort the perception of water availability, especially in regions with strong surface water-groundwater interactions.

In this context, an expeditious analysis conducted in a watershed with a surface water body and an unconfined aquifer indicated that water volumes discharged through surface and subsurface outlets were approximately equal (1:1). However, water transit times differ substantially: surface runoff responds almost immediately to precipitation, whereas groundwater flow exhibits a longer time lag. This difference, often ignored in static evaluations, may lead to misinterpretations of apparent water balance imbalances when the system is actually in dynamic equilibrium. Models that do not integrate these multicomponent response times tend to overestimate seasonal deficits or surpluses and may generate false indications of hydrological imbalance.

In addition, a common limitation of many hydrological models is the static or over simplified treatment of soil moisture. In reality, soil moisture acts as a hydrological memory of the system, integrating the effects of climate, vegetation, and soil properties. Ignoring its temporal and spatial variability, especially in highly seasonal tropical climates, can compromise simulation (Seneviratne et al. 2010; Liu et al. 2020).

Soil moisture dynamics vary significantly according to land use: forested areas tend to exhibit greater infiltration and water retention, whereas agricultural soils show higher surface runoff and lower water storage capacity. In areas with evergreen vegetation and deep roots, topsoil moisture may appear stable while vertical water extraction and redistribution remain active at depth, which compromises the use of shallow sensors as the sole indicator of water status. In addition, soil moisture behavior at plot scale may not reflect emerging patterns at basin or regional scales, reinforcing the need for multiscale approaches.

An important limitation identified across the reviewed literature is the incomplete characterization of rooting depth in both field measurements and modeling studies. Many investigations assess only the upper soil profile, often limited to the first meter, despite evidence that roots in tropical and semi-arid ecosystems can access water several meters below the surface. This underrepresentation can substantially overestimate groundwater recharge by neglecting deep root water uptake from stored soil moisture or shallow

Table 9 Disregards in recharge estimate

Factor	Associated error
Exclusive use of simple water balance (P - ET)	Overestimation of recharge due to not considering storage and lateral runoff
Bypass hydraulic redistribution	Underestimation of surface moisture and vegetation-conditioned fluxes
Do not consider preferential flow (macropores)	Underestimation of recharge in structured soils, especially under forest vegetation
Ignore Groundwater Flow Between Basins	Underestimation of system output and overestimation of local recharge
Static treatment of soil moisture	Failure to detect recharge pulses and hydrological memory of the system
Lack of direct validation data	Unverified recharge estimates, with high uncertainties
Low spatial representativeness of the data	Extrapolation of point data to heterogeneous areas

Values compiled from available comparable studies; sample size varied by category

groundwater. The problem is particularly critical in semi-arid environments, where relatively small differences in effective rooting depth may produce large changes in estimated recharge. These findings reinforce the need for deeper root profiling and explicit representation of rooting depth in recharge assessments.

Despite recent advances, significant knowledge gaps persist. Few field studies simultaneously evaluate root system behavior, plant water uptake depth, hydraulic redistribution, and vertical and lateral water flows within the soil profile. There is also a lack of continuous long-term datasets that allow the assessment of the seasonality of evapotranspiration and recharge under different land uses and edapho-climatic conditions. The absence of direct validation of recharge through storage variations, environmental tracers, or groundwater monitoring prevents clear distinction between deep infiltration and effective recharge (Table 9). In this context, the integration of experimental approaches, remote sensing, and distributed hydrological models is essential to overcome these limitations and improve the realistic representation of recharge processes in anthropized tropical areas.

Conclusions

This work shows that the presence of deep-rooted vegetation, particularly in forests and planted forest systems, significantly influences evapotranspiration (ET) dynamics and aquifer recharge. Evidence from the reviewed literature indicates that recharge under forest cover, in humid climates, tends to be 10% to 30% lower than that observed in pastures, due to the increase in ET losses, and may be reduced by up to 70% in semi-arid climates due to the direct

groundwater uptake by roots. This effect is particularly evident in mature forests or planted forests in tropical regions with deep soils. In temperate climates, reported differences between forest and pasture recharge are generally smaller, typically ranging from 5 to 15%.

Although part of the classical literature has associated forests with improved infiltration and increased recharge, more recent studies, including those analyzed in this review, indicate that this effect is not universal. Recharge magnitude depends on a set of interdependent factors, such as soil depth, vegetation type, land use management, climate, and root characteristics. Thus, the direct association between forest and higher recharge should be analyzed with caution, especially in strongly modified tropical environments. The main contribution of this review is to show that forest effects on recharge are context-dependent rather than universally beneficial or detrimental.

At the same time, other studies highlight compensatory ecohydrological mechanisms, such as hydraulic redistribution and greater vertical connectivity promoted by root macropores, which may enhance recharge under certain conditions even in areas of high ET. Agroforestry systems can combine agricultural crops with tree species, promoting deeper, more diverse root structures that improve water infiltration and reduce soil erosion. Conservation practices such as no-tillage help preserve soil cover and minimize soil compaction, thereby maintaining soil moisture and the porous structure required for infiltration. These practices reduce water losses through surface runoff and promote percolation into deeper soil layers, thereby helping sustain aquifer recharge over time. Although promising, these findings still require broader empirical validation across different tropical biomes.

Overall, the results emphasize the need to integrate soil conservation practices and sustainable land-use planning to optimize aquifer recharge, minimize negative impacts on the hydrological cycle, and improve subsurface water storage management, particularly in water-scarce regions. Further, local studies are needed to understand the specific dynamics of tree vegetation within watershed water balance, especially under climate change and increasing groundwater demand. Major knowledge gaps remain regarding the long-term effects of root depth, species composition, and land use transitions on recharge across contrasting tropical soils and climates. In addition, more integrated field monitoring, remote sensing, and coupled surface–subsurface modeling approaches are needed to distinguish transient soil water storage from effective aquifer recharge. Addressing these uncertainties may support more robust groundwater management strategies under future climate and land use change scenarios.

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Data availability All data presented in graphical form can be provided if requested.

Declarations

Ethics approval All authors have read, understood, and complied as applicable with the statement on “Ethical responsibilities of Authors” as found in the Instructions for Authors.

Consent to Participate This study did not involve human participants, and consent to participate was not required.

Competing interests The authors declare no competing interests.

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